

Air Quality RapidMiner Predictive Modeling Report by Vamshi Aitharaju

Introduction

For the Rapid Miner Predictive modeling, Initially I thought of forecasting the air pollution by years, but the model was not good and could not get what I wanted to show. So, I have proceeded to predict the values using Decision tree and Random Forest. So, I started with linear regression for Air pollutant value and pollutants and years. The model did not work, values were not up to the mark, and I could not get anything through it. so, after brainstorming, I thought as I could not use years for prediction, maybe I can predict values of pollutants by season and place (community District). So that's how I have chosen these variables like by playing with all the variables I got it question of predicting pollutant by seasons and place.

An Explanation of How This Adds Value to Decision Making

Why these models? As the forecasting model was not giving the results I want, I have chosen this Linear regression, decision tree, Random Forest for prediction. This model will definitely add value to my decision making as with models with selected variables, machine learning, I can identify not only factors that are affecting air quality but also how much they are influencing the air quality, which can be helpful to make environmental planning, policies and public health decisions. In simple words, Linear regression tells us quantitative evidence for planning, Decision Tree provides simple interpretation for people who is going make decision on this data, Random forest for reliable forecasting for future decisions.

Approach and methods used for predictive modeling

I have used rapid miner to build and compare predictive models that help in predicting air pollutant levels across the new York city. My main goal was to use machine learning to predict the Data Value (pollution concentration) by using Season, Pollutant Name, and Geo Place Type (Community District) as the input fields.

For this firstly I have prepared the data for rapid miner, I want all of my data to be in single sheet in a way that rapid miner find easy to read. Then I have retrieved the data in rapid miner, then I have used the operator Select Attribute, to select the attributes, so as per my business question I have selected the Data value(pollution level), Name(NO2,PM2.5), Geo place type and Season. I have selected only this attributes with which we can make my model efficiently. Then I have used Split data operator for into two parts, 70% for training and 30% for the testing just to make sure that how it perform on my data. Then I have started building my models. First, I used the Decision Tree operator with the Gini Index. This helped me to make a simple tree that shows how pollutant, season, and district affect the pollution value. Then I used the Linear Regression operator to check the numeric relation between inputs and output. It helped me to understand how much the pollution level changes with season or district. After that, I used the Random Forest operator to make a stronger model. I kept the number of trees as 100, and it used bootstrap sampling to take random data and make predictions more accurate. I compared all

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three models based on their performance and accuracy. All models helped me to understand the seasonal pattern of PM_{2.5} and NO₂ and which factors are more important for air quality prediction. Then I have done model performance, After running all my models, I compared them using R², RMSE. The Random Forest model gave the best accuracy, and the Decision Tree was easiest to explain. I kept all three because each model helped me in a different way.

A Summary of What My Model Predicts

In my RapidMiner project, I used Season, Pollutant Name, and Geo Place Type as my input columns. These are the main factors that affect air pollution. My output is Data Value, which shows the pollution level for PM_{2.5} and NO₂.

In my **Decision Tree model**, I saw that the decision tree is different for NO₂ and PM_{2.5}. The model first split by Pollutant Name, then by Season, and then by Community District. For NO₂, it focused more on Season and showed higher values in Winter. For PM_{2.5}, it focused more on District because pm_{2.5} stays almost steady across seasons. The R² was 0.784, which means it explained about 78% of the changes in pollution.

In my **Linear Regression model**, I used the same inputs to find the numeric relation between Season, Pollutant, and District. The R² was 0.692, meaning it explained around 69% of the variation. It showed that NO₂ increases more in Winter, while PM_{2.5} stays between 8 to 11 µg/m³. This model helped me see the seasonal and location effects in clear numbers.

In my **Random Forest model**, The R² was 0.782, almost same as Decision Tree. It split first by Pollutant, then Season, and then District. For NO₂, the model predicted higher values in Winter (around 20–27 ppb), while PM_{2.5} stayed between 8–11 µg/m³.

Model 1 and Model 5 gave the best results. Model 1 split by Pollutant and Season, showing NO₂ high in Winter and PM_{2.5} steady. Model 5 started with District and then Pollutant. Both models proved that Season affects NO₂ more, while PM_{2.5} changes slightly by District.

Conclusion based on my models and results:

All my models predicted the same pattern that NO₂ levels are higher during Winter, while PM_{2.5} remains steady across all seasons. Season has the biggest effect on pollution, followed by Pollutant and then District. The results show that models I choose can accurately predict future pollution values. Based on previous models I made and this model I can create dashboards on basis of place, seasons and Air pollutants.

Hereby I, Vamshi Aitharaju, declare that this models and report was completely done by me and ready for review. Thank You